KRIPAKARA M. N.

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Mangaluru, Karnataka(Open to relocate) • Portfolio

PROFESSIONAL SUMMARY

AI and full-stack developer with internship experience at Samsung R&D and CodeIO. Built an NLP ensemble reaching 98.48% accuracy for voice-command detection and a full-stack platform serving 5,500+ students. Skilled in Python, React, Node.js, and multi-agent LLM systems (LangGraph, MCP), with a focus on production-minded engineering

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript (ES6+), TypeScript, SQL
Web Development: React.js, Node.js, Express.js, REST APIs, FastAPI, HTML5, CSS3
Databases: MongoDB, PostgreSQL, Firebase (Auth, Firestore)
AI/ML: LLM Integration, RAG, Multi-Agent Systems, LangGraph, PyTorch, Hugging Face, LangSmith, NLP
Tools and Cloud: Git, Docker, AWS, Postman, Cursor

EXPERIENCE

Software Development Intern, CodeIO Bengaluru, India Mar 2024 – May 2024

- Developed a full-stack exam management platform **(React.js, Node.js, MongoDB)** serving 5,500+ students.
- **Automated exam scheduling, seating allocation, and hall-ticket generation** (PDFKit**,** Nodemailer) with a capacity and batch-aware allocation algorithm, **cutting administrative effort by ~60%.**
- **Implemented role-based access control (RBAC)** to isolate workflows and permissions for admins, invigilators, and students.

Samsung Research Intern, Samsung R&D Institute India, Bengaluru Nov 2022 – Aug 2023

- **Improved voice-command detection accuracy to 98.48% (98.46% F1)** with a 3-model NLP ensemble (**ALBERT, scikit-learn Naive Bayes, rule-based classifier)** using PyTorch, outperforming standalone BERT by ~2.5% pts.
- **Optimised inference latency to 56ms (detection) and 24ms (segmentation)** by curating targeted datasets with Pandas and building a lightweight 46MB ALBERT-based ensemble.
- Built a Gradio interface for real-time validation and testing of the model.

PROJECTS

ContractBreaker AI Reliability Testing Tool | Python, AST, pytest, | Finalist, Alan AI OpenFire Hackathon (Oct 2026)

- Engineered an open-source reliability testing tool to expose duplicate payment risks in Python HTTP clients through controlled failure simulation and retry analysis.
- Instituted deterministic failure verification using execution traces, integrated Alan AI through a Skill-based approval workflow, and validated the system with **319 tests and 42 benchmark cases**.

BillaraAI Smart Invoice Generator | React, TypeScript, Firebase (Auth & Firestore), Resend API

- Engineered a **full-stack AI invoicing platform (Gemini)** that converts text and audio briefs into structured invoices in seconds, cutting creation time by ~65% vs. manual entry.
- Architected a secure backend with **server-side API proxying**, Firestore RBAC rules, and **rate limiting**.
- Developed a real-time dashboard with penalty calculation, email dispatch, and PDF export.

LiveDataViz Conversational Data Visualization | React.js, JavaScript (ES6+), RAG, BM25

- **Reduced chart creation time by ~80%** by developing an **NL2Code** engine that translates user queries into structured Vega-Lite schemas.
- **Built a hybrid RAG** (**Gemini Embeddings** + **BM25)** that retrieves dataset statistics, categorical values, and sample records to ground LLM-generated charts in the uploaded data.
- **Designed a vector store** with automatic 30-minute TTL cleanup and explicit session deletion.

TripBuddy AI Multi-Agent Travel Planner | Python, Next.js, LangSmith, FastAPI, PostgreSQL, Docker

- Orchestrated a **5-agent LangGraph system across 9 workflow nodes** for flight, hotel, weather, budgeting, and itinerary planning with supervisor-based routing.
- Integrated **3 live data providers** via **Model Context Protocol (MCP)** over remote HTTP and local stdio, and added a human-in-the-loop approval step for critical booking decisions.
- Containerized the app with a multi-stage **Docker build (non-root user, health check)** and deployed it.

EDUCATION

B.M.S College of Engineering, Bengaluru Dec 2020 – Jun 2024
B.E. in Computer Science and Engineering, CGPA: 8.2/10